33. (a) (i) Explain the phenomenon of total internal reflection. State two conditions that must be satisfied for total internal reflection to occur. Derive the relation between the critical angle and the refractive index of a medium.

(ii) A ray of light is travelling from a denser medium of refractive index 1·5 to a rarer medium of refractive index 1·0. Calculate critical angle for total internal reflection. CBSE 26C 5

(i) When a ray of light enters from a denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases at the interface, angle of refraction also increases till the angle of incidence is $\frac{\pi}{2}$ (i.e the refracted ray grazes the surface of the interface). If the angle of incidence is further increased, the incident ray totally gets reflected back. This is called total internal reflection. | 2

Conditions

1. Light must travel from a denser medium to a rarer medium. | ½

2. At the interface, the angle of incidence must be greater than the critical angle. | ½

Relation

When light goes from a denser medium (refractive Index n_2) towards a rarer medium (refractive Index n_2), then for the value of critical angle, we have relation-

$n_2 \sin i_c = n_1 \sin 90^\circ$ | ½

$\sin i_c = \frac{n_1}{n_2}$

or, $\frac{n_2}{n_1} = \frac{1}{\sin i_c}$

or,

$n_{21} = \frac{1}{\sin i_c}$ | ½

(ii) Calculating critical angle

$n_2 = 1.5 = \frac{3}{2}$, $n_1 = 1$

$\sin i_c = \frac{1}{n_{21}} = \frac{2}{3}$ | ½

Therefore,

$i_c = \sin^{-1}\left(\frac{2}{3}\right)$ | ½

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0° (B) 30°

(C) 60° (D) 90°

CBSE 26C

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0° (B) 30°

(C) 60° (D) 90°

CBSE 26C

(A) 0^0

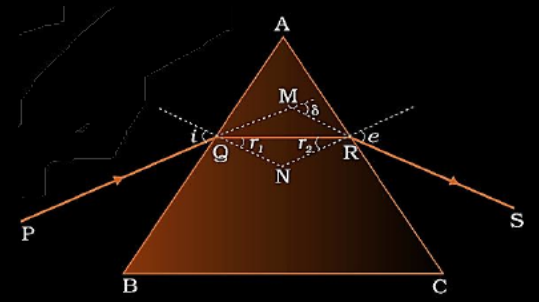
24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. *3*

CBSE 26C

24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. 3

CBSE 26C

In quadrilateral AQNR,



$\angle A + \angle QNR = 180°$

From triangle QNR,

½

$r_1 + r_2 + \angle QNR = 180°$

Comparing the above equations,

$r_1 + r_2 = A$

The total deviation δ is the sum of deviations at the two faces:

$\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - A$ ½

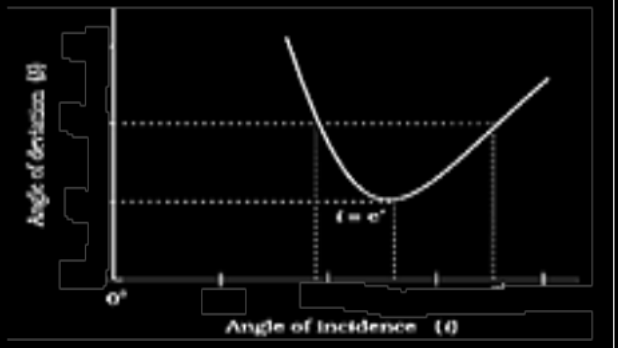
At minimum deviation,

$\delta = D_m$, $i = e$ *which implies* $r_1 = r_2$

$r = \frac{A}{2}$

$D_m = 2i - A$ or $i = \frac{A+D_m}{2}$ ½

The refractive index of the prism is



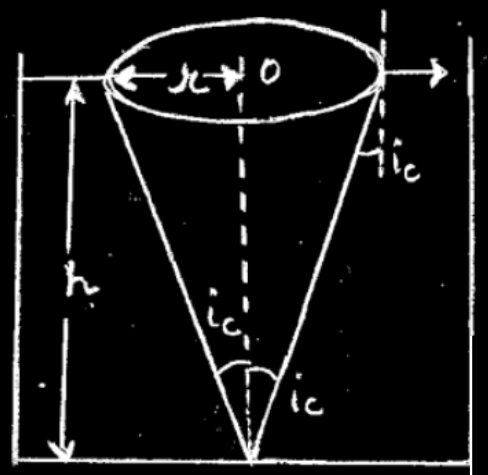
$n = \frac{\sin i}{\sin r}$ 1

$n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ ½

33. (a) A small bulb is placed at the bottom of a tank containing water to a depth of 1·00 m. Find the area of the surface of water through which light from the bulb can emerge from. The refractive index of water is (4/3). *3*

(b) An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed coaxially at a distance of 10 cm from the lens. If the image formed by this combination coincides with the object itself, draw the ray diagram and find the focal length of the convex mirror. *2*

CBSE 26C



1

$\sin i_c = \frac{1}{n}$ ½

$\tan i_c = \frac{r}{h}$

$r = h \tan i_c$ & $\tan i_c = \frac{1}{\sqrt{n^2-1}}$

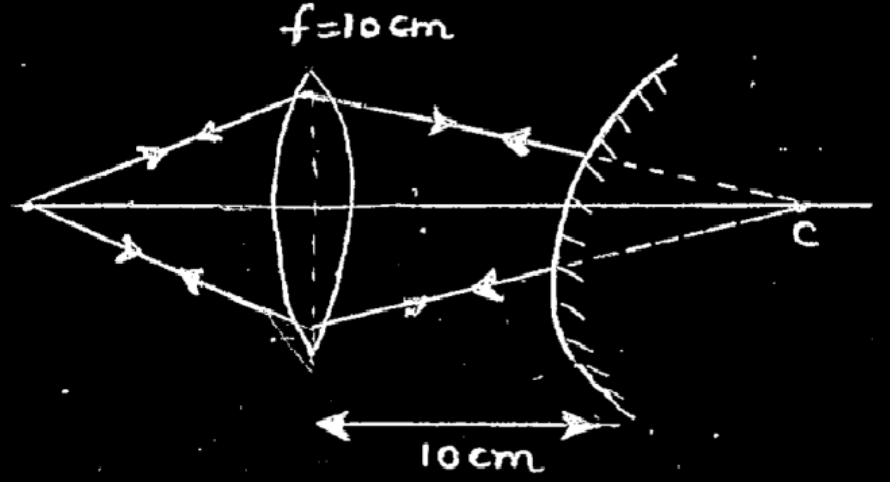
$A = \pi r^2 = \pi h^2 \tan^2 i_c$ ½

$A = \frac{\pi h^2}{n^2-1}$ ½

$A = \frac{\pi \times 1}{\frac{16}{9}-1}$ ½

$= \frac{9\pi}{7}$ m^2 $= \frac{9\times 3.14}{7}$ =4.04 m^2

(b)



1

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{15}$

$= \frac{1}{30}$

$v = 30$ cm ½

Radius of curvature of the mirror,

$R = 30 - 10 = 20$ cm

$f = \frac{R}{2}$ ½

$f = 10$ cm

24. (a) The distance of an object from a concave mirror is increased from 12 cm to 15 cm. It is observed that the new image becomes $\left(\frac{2}{5}\right)^{\text{th}}$ the size of the initial image. Calculate the focal length of the mirror.

(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C *3*

(a)

For $u = -12\ cm$

$$m_1 = \frac{h_1}{h_o} = \frac{f}{f+12} \quad \ldots(1)$$ ½

For $u = -15\ cm$

$$m_2 = \frac{h_2}{h_o} = \frac{f}{f+15} \quad \ldots(2)$$ ½

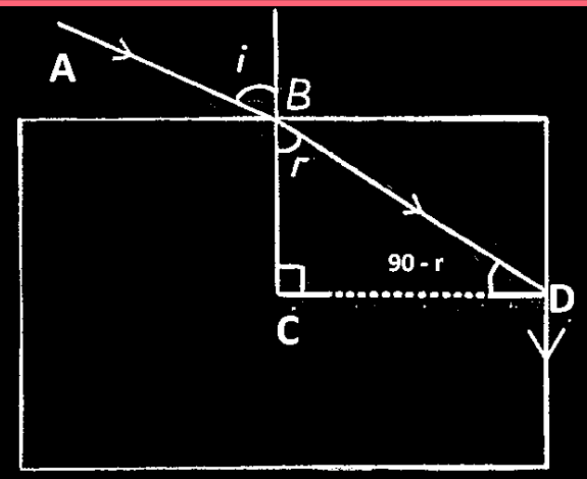
Dividing (1) by (2)

$$\frac{m_1}{m_2} = \frac{h_1}{h_2} = \frac{f+15}{f+12}$$

$$h_2 = h_1\left(\frac{2}{3}\right)$$

On solving,

$$f = -10\ cm$$ ½



At air-glass interface,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin r}$$

$$\sin r = \frac{\sin 45^\circ}{n} = \frac{1}{\sqrt{2}\,n} \quad \ldots(1)$$ ½

At glass-air interface, light suffers Total Internal Reflection (TIR) ($i_C = 90°\text{-}r$).

$$n = \frac{1}{\sin i_C}$$

$$n = \frac{1}{\sin(90^\circ - r)} = \frac{1}{\cos r}$$

$$n = \frac{1}{cosr}$$

$$\cos r = \frac{1}{n} \quad \ldots(2)$$ ½

Squaring and adding (1) and (2)

$$1 = \frac{1}{2n^2} + \frac{1}{n^2}$$

On solving,

$$n = \sqrt{\frac{3}{2}}$$ ½

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

(B) π/4

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

$$n = \frac{\text{speed of light in vacuum(c)}}{\text{speed of light in medium(v)}} \quad \ldots\ldots\ (i)$$ ½

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$ ½

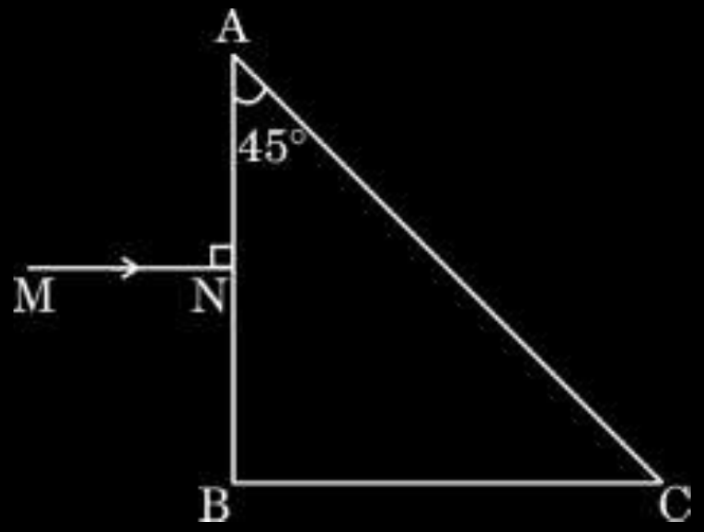
$$n = \frac{12.5}{9} \quad \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (ii)$$

From (i) and (ii)

$$v = \frac{3 \times 10^8 \times 9}{12.5}$$ ½

$$= 2.16 \times 10^8 \text{ m/s}$$ ½

18. A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$. *2*



CBSE 2026

(i)

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$$

$$i_c = 45^0$$

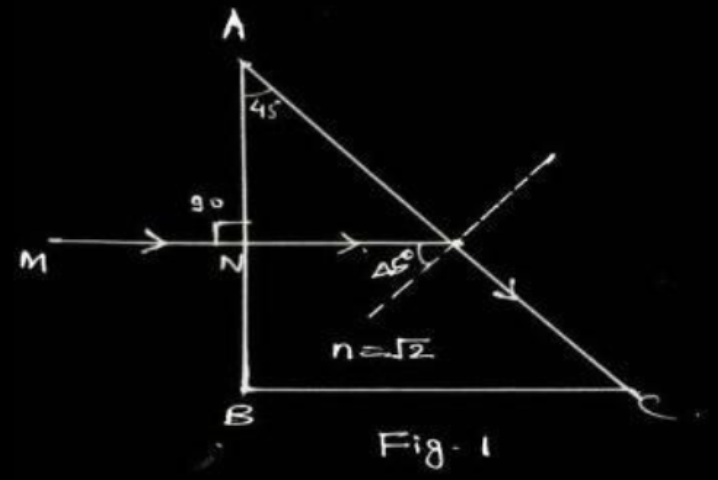


Fig. 1

1

Since $i = i_c$ so the emergent ray grazes alongside AC.

(ii)

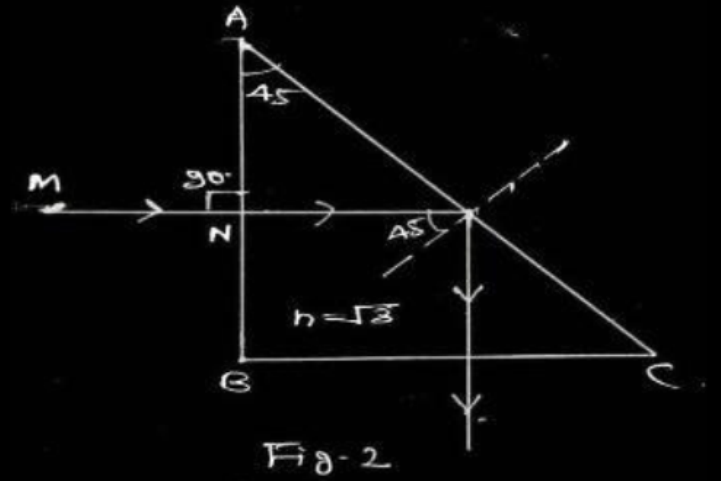


Fig. 2

1

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$$

$$i_c < 45^0$$

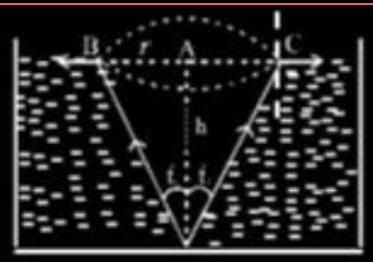
Since $i > i_c$ so, the ray suffers from Total Internal Reflection (TIR)

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

CBSE 2026

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

CBSE 2026



$r = h \times \tan i \quad [\text{as } i = i_c]$ ½

$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}} \Rightarrow i_c = 45^\circ$ ½

$r = h$

$\text{Area} = \pi r^2$ ½

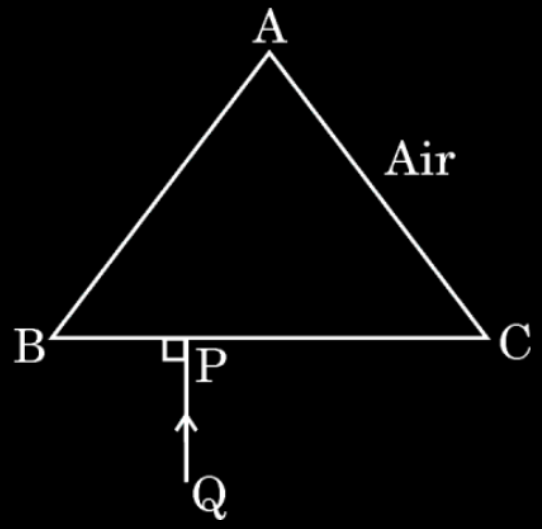
$= 3.14 \times (1)^2$

$\text{Area} = 3.14\ m^2$ ½

31. (a) (i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum derivation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation. *5*



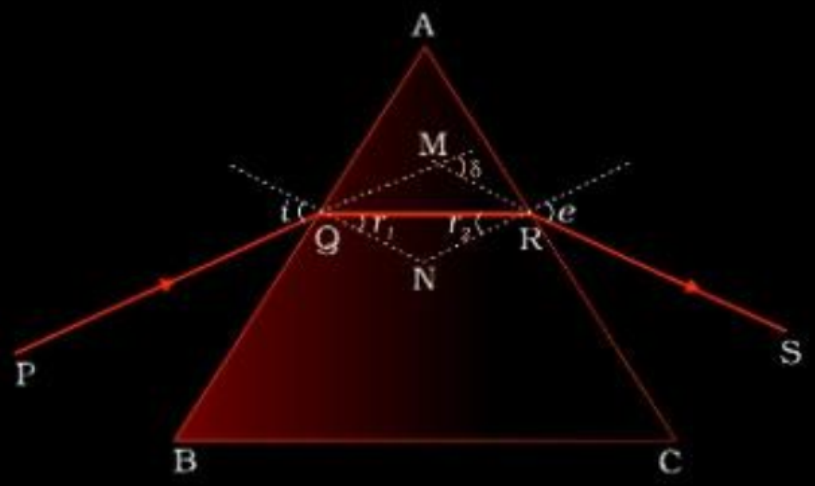
CBSE 2026

(i) Refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium. — 1

Alternatively

$$n_{21} = \frac{c}{v}$$

(ii)



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180°.

$\angle A + \angle QNR = 180°$

From the triangle QNR,

$r_1 + r_2 + \angle QNR = 180°$

Comparing these two equations, we get

$r_1 + r_2 = A$ .....(i) — ½

The total deviation δ is the sum of the deviations at the two faces.

$\delta = i + e - A$ .....(ii) — ½

For $\delta = \delta_m$; i = e which implies $r_1 = r_2$

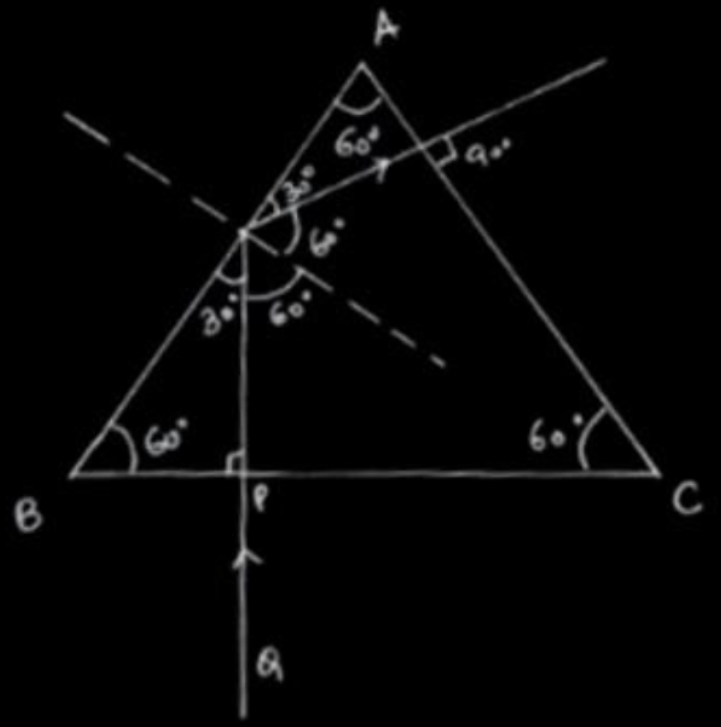
From equation (i) and (ii) — ½

$$r = \frac{A}{2}; i = \frac{A + \delta_m}{2}$$

Refractive index of the prism

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\frac{(A + \delta_m)}{2}}{\sin\frac{A}{2}}$$ — ½

(iii)

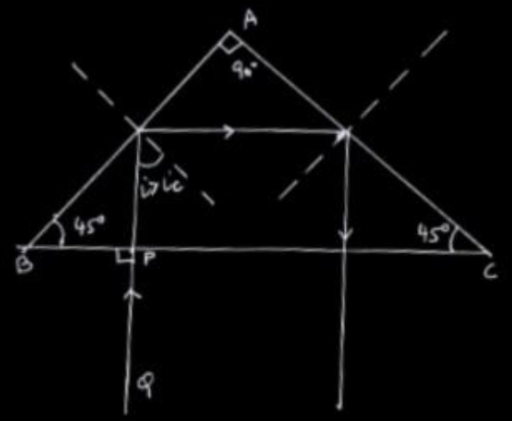


1

The ray QP strikes the face BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. It strikes the face AC normally and comes out undeviated.

1

Alternatively



The ray QP strikes the surface BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. As at face AC, $i > i_c$, it undergoes TIR. The ray comes out of the face BC undeviated.

(Award full credit of this part if the student takes the value of A other than 60° or 90° and traces the path correctly and according to the angle taken)

OR

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

(A) Twinkling of stars

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

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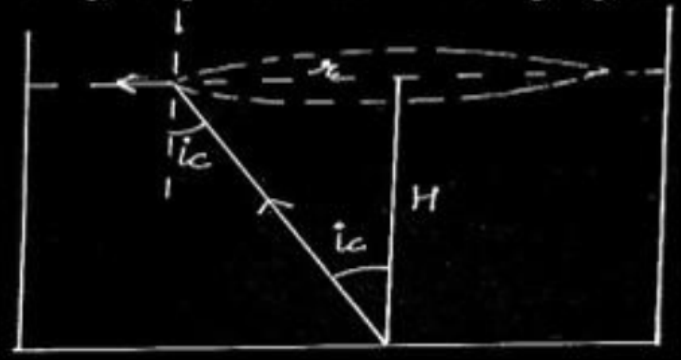
26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

CBSE 2026

$\sin i_C = \frac{1}{n}$	
$= \frac{2}{3}$	1
For glass – water surface	
$\frac{3}{2}\sin i_C = \frac{4}{3}\sin r$	½
$\frac{3}{2} \times \frac{2}{3} = \frac{4}{3}\sin r$	
$\sin r = \frac{3}{4}$	
For water – air surface.	½
$\frac{4}{3}\sin r = \sin e$	
$\frac{4}{3} \times \frac{3}{4} = \sin e$	½
$\sin e = 1$	
$e = 90^\circ$	½

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

Critical angle: The angle of incidence corresponding to the angle of refraction 90° for a given pair of media when light goes from denser to rarer medium.	1
	½
$\text{Sin } i_c = \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + H^2}}$	½
$\mu^2 r^2 = r^2 + H^2$	½
$(\mu^2 - 1)r^2 = H^2$	
$r = \frac{H}{\sqrt{\mu^2 - 1}}$	½

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

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9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

(C) Total internal reflection occurs for both green and blue lights.

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

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23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

(a) As ray of light passes symmetrically through the prism

$\angle r_1 = \angle r_2 = r$, $i = e$ ½

$\delta_m = 2i - A$ ½

$= 2 \times 45° - 60°$

$= 30°$ ½

(b) $r = \frac{A}{2} = 30°$

$\mu = \sin \frac{(\frac{A + \delta_m}{2})}{sin\frac{A}{2}}$ ½

$= \sin \frac{(\frac{60° + 30°}{2})}{sin30°}$ ½

$= \frac{sin45°}{sin30°}$

$= \sqrt{2}$ ½

(Alternatively)

$\mu_a \sin i = \mu_g \sin r$

$1 \times \sin 45° = \mu_g \times \sin 30°$

$\mu_g = \sqrt{2}$

4. The process that allows light to travel through a tube which is not straight is – 1

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

4. The process that allows light to travel through a tube which is not straight is – 1

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

(D) total internal reflection of light

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

Angle of incidence $i = \frac{3}{4}A = 45^o$	1/2
At minimum deviation $r = \frac{A}{2} = 30^o$	1/2
$n = \frac{\sin i}{\sin r} = \frac{\sin 45^o}{\sin 30^o}$	
$n = \sqrt{2}$	1/2
$n = \frac{c}{v}$	
$v = \frac{3}{\sqrt{2}} \times 10^8$ m/s = 2.12×10^8m/s	1/2

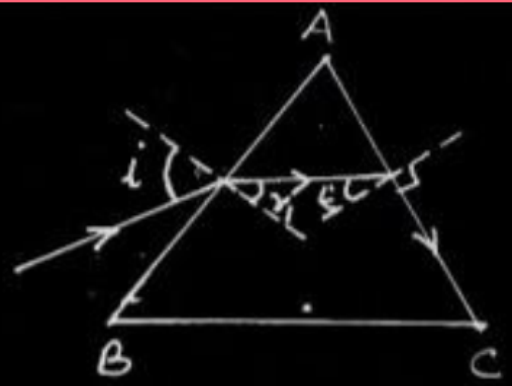
19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

$\sqrt{2}(\sin i_c) = 1$ — ½

$(\sin i_c) = \frac{1}{\sqrt{2}}$

$i_c = 45^0$



$r_1 + r_2 = A$ (Here $r_2 = i_c$) — ½

$r_1 = A - i_c$

$= 60^0 - 45^0$

$= 15^0$ — ½

$\text{Sin } i = \sqrt{2}\sin 15^0$

$i = \sin^{-1}[\sqrt{2}\sin 15^0] = \sin^{-1}(0.366)$ — 1/2

21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

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21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

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1

From Snell's Law

$n_1 . \sin i = n_2 . sinr$ ½

$A = r_1 + r_2$

$A = r_1 \quad \because r_2 = 0$

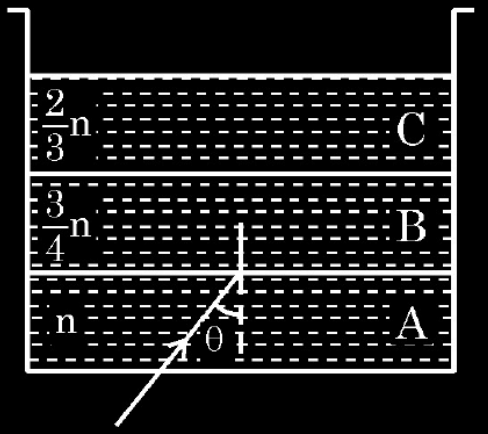
$1 \times sin2A = n \times sinA$

$2\ sinA\ cosA = n \times sinA$

$n = 2cosA$ ½

28. (a) State the two conditions under which total internal reflection occurs. **3**

(b)



A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n, $\frac{3}{4}$ n and $\frac{2}{3}$ n, respectively. A laser beam is incident at the interface between A and B at an angle θ as shown in figure. Prove that the beam does not enter region C at all for $\sin\theta \geq \frac{2}{3}$.

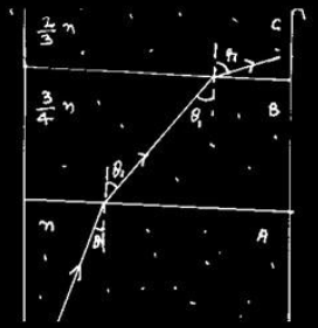
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(a) (i) Light must travel from denser medium to rarer medium. ½

(ii) Angle of incidence must be greater than Critical angle. ½

$i > i_c$

(b)



For refraction at AB interface

$$\frac{\sin\theta}{\sin\theta_1} = \frac{3n/4}{n}$$

$$\sin\theta = \frac{3}{4}\sin\theta_1 \quad \text{---------- (1)}$$ ½

For refraction at BC interface

$$\frac{\sin\theta_1}{\sin r} = \frac{2n/3}{3n/4}$$

$$\sin\theta_1 = \frac{8}{9}\sin r \quad \text{---------- (2)}$$ ½

From equations (1) and (2)

$$\frac{3}{2}\sin\theta = \sin r$$ ½

for $\sin\theta = \frac{2}{3}$

$\sin r = 1$ i.e $r = 90^0$ ½

Ray grazes the surface

Hence for $\sin\theta \geq \frac{2}{3}$ does not enter the region C at all